

Vitamin B2 (Riboflavin)

Dosage & Migraine Guide

The migraine-prevention dose used in clinical trials, the two forms of B2 available, and who is actually at risk of deficiency.

Evidence-based approach: Schoenen et al. (1998), *Neurology*; Condò et al. (2009), *The Journal of Headache and Pain*.

Why It Works

Riboflavin becomes FAD and FMN, coenzymes required by Complex I and Complex II of the mitochondrial electron transport chain. A 1998 randomized, double-blind, placebo-controlled trial by Schoenen and colleagues found that 400mg of riboflavin daily for three months cut migraine frequency by at least half in 59% of patients, versus 15% on placebo.

Dosing At A Glance

Goal	Dose	Notes
General adequacy (RDA)	1.1-1.3mg/day	Easily met through diet alone
Migraine prevention	400mg/day	Minimum 3-month trial before assessing
Pediatric migraine	200mg/day	Per Condo et al. 2009, under medical guidance
R5P bioactive form	50-100mg/day	Skips one liver conversion step

Who's Actually At Risk of Deficiency

- Chronic alcohol use — impairs riboflavin absorption and hepatic storage.
- Long-term oral contraceptive or tricyclic antidepressant use — interferes with riboflavin metabolism.
- Strict vegan diets without fortified foods.
- Endurance athletes — riboflavin turnover rises with training volume.

Quality Checklist

- ☐ Single-ingredient product, not buried in a proprietary blend
- ☐ Third-party tested (USP, NSF, or Informed Choice)
- ☐ Labeled milligram amount matches what you intend to dose
- ☐ Expect bright yellow urine at high doses — harmless and expected

Sources

Schoenen, J. et al. (1998). Effectiveness of high-dose riboflavin in migraine prophylaxis. *Neurology*, 50(2), 466-470.

Condo, M. et al. (2009). Riboflavin prophylaxis in pediatric and adolescent migraine. *J. Headache and Pain*, 10(5), 361-365.

Canadian Headache Society (2012). Guideline for migraine prophylaxis. *Can J Neurol Sci*, 39(2 Suppl 2), S1-S59.